AI Powered Interview Evaluation & Soft Skill Analysis Module

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Recruitment processes have rapidly transitioned into digital platforms due to the growing volume of job applications and the need for efficient talent acquisition. Traditional recruitment practices mainly rely on manual resume screening, keyword based filtering, and subjective interview evaluation. These approaches are inefficient, time consuming, and often fail to evaluate critical soft skills such as communication ability, emotional intelligence, and behavioral competence. As a result, organizations may overlook suitable candidates while spending excessive time and resources during hiring processes and it leads to incomplete candidate evaluation and may result in unsuitable hiring decisions. Existing Applicant Tracking Systems (ATS) rarely incorporate automated interview analysis or soft skill assessment mechanisms.

The identified research gap lies in the lack of intelligent systems capable of evaluating candidate performance during interviews using multimodal analysis such as speech patterns, facial expressions, and behavioral cues. Current approaches mainly evaluate structured data such as resumes and job descriptions, while human recruiters must manually interpret interview responses, making the process subjective, time consuming, and potentially biased.

This research proposes an AI based interview evaluation module integrated within an AI Powered Intelligent Talent Acquisition System. The module analyzes candidate responses using Natural Language Processing (NLP), speech analysis, and emotion recognition techniques to evaluate communication skills, confidence, clarity, sentiment, and behavioral indicators. Machine learning models will be applied to interpret linguistic patterns, while computer vision techniques analyze facial expressions and emotional cues. The system will generate structured evaluation reports and soft skill scores that support recruiters in decision making.

The implementation will utilize technologies such as Python, TensorFlow, PyTorch, OpenCV, speech processing libraries, and NLP frameworks like spaCy and Transformers. Data will be collected from simulated interview sessions and public datasets while ensuring ethical data handling and privacy protection.

Expected outcomes include improved interview evaluation consistency, enhanced transparency in candidate assessment, and reduced recruiter workload. Socially, the system can promote fairer hiring practices by standardizing soft skill evaluation. Commercially, the solution can support organizations, especially small and medium enterprises, adopting intelligent and affordable recruitment technologies.

Keywords: Artificial Intelligence, Interview Analysis, Soft Skill Evaluation, Natural Language Processing, Emotion Recognition.

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LIST OF ABBREVIATIONS

Table 1 : List of Abbreviations

AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning
ATS	Applicant Tracking System
STT	Speech to Text
LLM	Large Language Model
CNN	Convolutional Neural Network
HR	Human Resources
PDF	Portable Document Format
SDG	Sustainable Development Goal

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1. INTRODUCTION

1.1 Background and Context

Recruitment is an important organizational process. It directly affects the workforce quality, how well they work and how successful they are for long term organizational performance. These days companies use job portals and digital platforms to find new employees, and they get large number of job applications for a one single job opening. It gives people more chances to find work and it makes things harder for the people in charge of hiring and evaluating candidates. They must look through a lot of information about each person who applies. Traditional ways of hiring people, like looking at resumes and doing interviews, basic keyword based filtering systems and subjective interview assessments are not good enough. Therefor companies need to know if someone has both technical skills and soft skills as if they should work well with others.

This problem affects lot of people such as Human Resource professionals, the companies themselves and the people who are looking for jobs. Recruiters often feel overwhelmed when interviewing large number of applications for a single job vacancy because they must go through every detail of applications to select the best. This could cause a delay in hiring process. This can be unfair to some applicants. And companies may miss the chance to hire talented people because traditional systems do not properly show their real abilities. his situation can frustrate everyone involved.

There are numbers that show how big a problem this is. Around the world most employers think that things like being able to communicate with work in a team and being flexible are important for succeeding at work. However, a lot of the technology used for hiring focuses on the technical skills that people list on their resumes. It does not do a job of figuring out if someone is good at working with others or if they have the right personality for the job. In companies, where there might be thousands of people applying for one job, it is even harder for the people in charge of hiring to really get to know each person.

The systems that companies use now to manage job applications like the ones that track and sort resumes can help with some things. They can make it easier to store resumes and manage the hiring process. They are not perfect. They often rely on keywords, and it can be misleading. Recruiters also cannot directly observe a candidate’s behavior in interviews. Such as their

communication style or reactions to situations. Because of these hiring decisions often depend on personal judgment which may lead to bias.

New technologies, like AI and NLP can help solve some problems like these. These technologies can analyze things such as what people say in interviews, how they say it and how they feel. Also, they can give companies an idea of who someone is and how they might fit in at the company. By using these technologies companies can make the hiring process fairer and more based on facts.

To deal with these issues this project is proposing a system that uses artificial intelligence to help evaluate job interviews and figure out if someone has the right soft skills. The system can look at what people say in interviews. How are there soft skills. This will help companies make better hiring decisions that are based on more than a resume. This project helps promote fair and inclusive hiring. It supports equal job opportunities by using AI to make the recruitment process more transparent and reduce bias. It ensures that every candidate has a fair chance to get a job.

From a research standpoint this project fits into the area of intelligence and data science especially when it comes to using machine learning and data to help companies make better decisions. The project adds to this area of research by looking at how artificial intelligence can be used to analyze job interviews and figure out if someone has soft skills. Recruitment is a part of this, and the project wants to make it better. The goal is to make the hiring process fairer and more effective and to help companies find people for the job.

1.2 Literature Review

The rapid advancement of digital technologies has significantly transformed the recruitment and hiring landscape. Traditionally, recruitment relied heavily on manual resume screening, recruiter intuition, and face to face interviews to assess candidate suitability. However, the emergence of Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) has enabled organizations to automate and enhance various aspects of the recruitment process. Modern AI driven recruitment systems aim to improve efficiency, reduce hiring time, and provide more objective evaluation of candidate capabilities.

Recent research highlights the increasing adoption of AI powered interview systems that utilize advanced computational techniques to analyze candidate responses, communication patterns, and behavioral characteristics. These systems attempt to simulate human like interview experiences by combining multiple technologies such as speech recognition, sentiment analysis, and computer vision. For example, Tighare et al. [1] introduced the Skill Sight interview assessment platform, which focuses on evaluating candidates' emotional intelligence by analyzing facial expressions, reaction time, and behavioral responses during interviews. The study emphasizes that nonverbal communication cues such as facial expressions and eye movements provide valuable indicators of a candidate's emotional awareness and confidence during professional interactions.

Similarly, Jagtap et al. [4] and Patil et al. [8] proposed AI driven interview simulation platforms that utilize real time voice based agents to conduct automated interview sessions. These systems employ Speech to Text (STT) technologies combined with conversational AI models to interpret candidate responses and generate follow up questions dynamically. Such systems attempt to replicate natural conversation patterns, thereby creating a more interactive and realistic interview experience. These frameworks also enable automated response evaluation by analyzing linguistic features such as vocabulary usage, sentence complexity, and contextual relevance.

Several studies have also explored the integration of deep learning techniques for candidate behavior analysis. Emotion recognition models based on Convolutional Neural Networks (CNNs) have been widely used to analyze facial expressions and emotional states during video interviews. These models can identify emotions such as confidence, nervousness, or uncertainty by analyzing facial landmarks and micro expressions [5]. In addition, advanced machine learning models such as XGBoost classifiers and Transformer based architectures have been applied to evaluate candidate responses and generate structured feedback for recruiters [10]. These approaches enable the extraction of meaningful patterns from candidate responses and allow systems to evaluate interview performance more systematically.

Furthermore, some recent research has explored the concept of immersive AI interview environments using advanced human computer interaction technologies. Wu et al. [7] proposed an AI interview evaluation framework that incorporates Virtual Digital Humans (VDH) to simulate human interviewers. These digital avatars interact with candidates in real time and provide a more engaging and realistic interview environment. The use of multi agent systems within such platforms allows dynamic question generation, behavioral analysis, and adaptive feedback mechanisms. These approaches demonstrate the potential of integrating AI driven conversational agents with advanced visualization technologies to create more interactive recruitment systems.

In addition to academic research, several commercial platforms have emerged that aim to modernize recruitment processes. Platforms such as HireVue, Spark Hire, and VidCruiter provide video interviewing solutions that allow recruiters to conduct remote interviews and record candidate responses for later evaluation [1]. These platforms integrate basic Applicant Tracking System (ATS) functionality, enabling organizations to store candidate data, manage interview scheduling, and track recruitment workflows. Similarly, specialized platforms such as Codility focus on technical skill assessment by providing automated coding tests and performance evaluation tools. Another example is Google's Interview Warmup, which is designed to help candidates practice interview questions and receive automated feedback on their responses.

Despite these technological advancements, current recruitment systems still exhibit several limitations. Many existing platforms primarily focus on technical skill assessment or video recording capabilities, without performing deep behavioral analysis of candidate responses. According to systematic reviews conducted by Sanjana et al. [2], [6], most AI driven recruitment systems lack integrated frameworks that simultaneously analyze linguistic content, emotional cues, and behavioral characteristics during interviews. As a result, these systems often provide limited insights into candidate communication abilities and interpersonal skills, which are critical for many professional roles.

Another key limitation identified in existing research is the lack of explainability and transparency in AI based candidate evaluation. In many automated recruitment systems, AI models generate candidate scores without providing clear explanations of how these scores were calculated. This lack of interpretability may lead to concerns regarding fairness, bias, and accountability in recruitment decisions [7], [10]. Furthermore, many platforms rely heavily on keyword based resume filtering techniques, which may overlook contextual skills and transferable experience [3], [9]. Such approaches may result in the rejection of highly qualified candidates who do not use specific keywords in their resumes.

Additionally, several studies highlight the absence of personalized feedback mechanisms within existing AI interview systems. While many platforms provide automated interview scoring, they rarely

offer detailed insights or learning recommendations that help candidates improve their performance [2], [6]. The absence of such pedagogical support limits the effectiveness of these systems as career development tools.

Overall, although current research demonstrates significant progress in AI powered recruitment technologies, existing systems still lack a holistic evaluation framework that integrates communication analysis, emotional intelligence detection, and explainable AI based scoring within a single platform. This limitation highlights the need for more comprehensive recruitment solutions that evaluate both technical competency and soft skills in an objective and transparent manner.

1.2 Research Gap

Despite the growing adoption of Artificial Intelligence in recruitment and talent acquisition systems, several significant limitations remain in current research and commercial solutions. While many existing systems incorporate AI technologies to automate resume screening or conduct basic interview simulations, most of these solutions focus on individual aspects of the recruitment process rather than providing a comprehensive evaluation framework. As a result, organizations still face challenges in accurately assessing candidate communication abilities, behavioral traits, and overall suitability for a given role.

One major research gap lies in the lack of integrated systems that combine multiple dimensions of interview analysis within a single framework. Most current recruitment technologies concentrate on either resume based candidate matching or technical skill testing, leaving interview performance largely dependent on subjective human judgment [1], [6]. Although video interview platforms allow recruiters to review recorded responses, they typically do not perform detailed automated analysis of candidate communication patterns, emotional expressions, or behavioral cues. This limitation prevents recruiters from obtaining deeper insights into a candidate's soft skill competencies, such as confidence, clarity of communication, and emotional intelligence.

Another limitation observed in current approaches is the overreliance on keyword based filtering mechanisms and rule based chatbot systems. Traditional Applicant Tracking Systems (ATS) primarily rely on matching keywords in resumes with job descriptions to shortlist candidates [3], [9]. While this approach may improve efficiency in large scale recruitment processes, it fails to capture contextual meaning, transferable skills, and deeper semantic relationships within candidate responses. Consequently, potentially qualified candidates may be rejected simply because their resumes do not contain specific keywords required by the system.

Additionally, many AI powered interview platforms lack advanced communication pattern analysis capabilities. Although some systems utilize Speech to Text technologies to transcribe candidate responses, these systems rarely analyze communication features such as speaking fluency, filler word usage, response structure, and articulation [5], [8]. These characteristics are important indicators of professional communication skills and can significantly influence interview outcomes in real world hiring scenarios.

A further research gap exists in the lack of explainable AI mechanisms within recruitment decision making systems. Many AI driven platforms provide a numerical score representing candidate performance without explaining how the score was generated or which factors influenced the evaluation. This lack of transparency may lead to algorithmic bias, reduced recruiter trust in AI

recommendations, and ethical concerns related to automated decision making [7], [10]. In modern AI systems, explainability is an important requirement to ensure fairness, accountability, and user acceptance.

Moreover, current systems often fail to incorporate behavioral and emotional analysis within interview evaluation processes. While some research has explored emotion recognition through facial expression analysis, the integration of emotional intelligence assessment with speech analysis and linguistic evaluation remains limited [2], [7]. As a result, existing systems cannot fully capture the behavioral and psychological aspects of candidate performance during interviews.

Another notable limitation is the absence of credential verification mechanisms within many AI recruitment platforms. Candidates often submit certificates and educational credentials as part of the hiring process, but manual verification of these documents can be time consuming and prone to human error. The integration of automated certificate validation within AI driven interview systems could significantly enhance trust and transparency in recruitment processes.

To address these limitations, this research proposes an AI Based Interview Evaluation and Soft Skill Analysis module that introduces a multi-dimensional assessment framework. Unlike traditional recruitment systems that focus primarily on resumes or technical testing, the proposed solution evaluates candidate interview performance using a combination of speech recognition, natural language processing, sentiment analysis, and communication pattern detection techniques. This approach enables the system to analyze both the content and delivery of candidate responses, providing deeper insights into their communication abilities and behavioral characteristics.

The proposed module also introduces an explainable scoring mechanism that generates detailed evaluation reports for recruiters, highlighting the factors contributing to each candidate's performance score. By combining linguistic analysis with behavioral indicators such as fluency, tone, and response structure, the system generates a comprehensive Candidate Readiness Profile that supports more informed hiring decisions.

Therefore, the novelty of this research lies in the development of a holistic AI driven interview evaluation framework that integrates communication focused soft skill analysis with credential validation and explainable scoring mechanisms. This integrated approach addresses the limitations of existing recruitment technologies and provides organizations with a transparent, objective, and scalable solution for modern talent acquisition processes [9], [10].

2. OBJECTIVE

2.1 Main Objective

To design and develop an AI driven interview evaluation module that automatically assesses a candidate's professional communication patterns and soft skill proficiency by analyzing verbal responses and vocal metadata to provide an objective, data driven readiness score.

SMART Analysis of the Main Objective

- Specific (S): The objective clearly defines the development of an AI system that evaluates candidate interviews, focusing on communication, behavioral cues, emotional intelligence, and credential validation. The system will use speech to text transcription (OpenAI Whisper), NLP for semantic analysis, and sentiment and emotion detection models to generate interpretable soft skill assessments and feedback for recruiters.

- Measurable (M): Success will be measured by:
    1. Achieving agreement with human HR evaluations for most of the soft skill and communication scores.
    2. Reducing the average interview evaluation time compared to manual assessment.
    3. Generating explainable AI reports that highlight reasoning behind scores.

- Achievable (A): The objective is achievable within a final year research project as it leverages existing AI models, frameworks, and APIs for speech recognition, NLP, and sentiment analysis. A functional prototype will be developed and validated using simulated and real interview datasets.

- Relevant (R): This objective directly addresses research gaps identified in the literature review, particularly the absence of real time, multi-dimensional soft skill evaluation, behavioral cue analysis, and explainable scoring mechanisms in recruitment. The system improves efficiency, fairness, and transparency in hiring decisions.

- Time bound (T): The project will be completed within 12 months, including stages for system design, model development, integration, testing, and final report submission by December 2026.

2.2 Specific Objectives

Objective 1: Develop a Speech to Text Interview Transcription Module

- **Specific:** Design and implement a transcription module using OpenAI Whisper to convert candidate interview audio into text with high accuracy.
- **Measurable:** Assess transcription correctness by verifying the preservation of meaning and exact wording across various speech patterns.
- **Achievable:** Existing speech recognition models can be fine-tuned and integrated for reliable transcription.
- **Relevant:** Accurate transcription is critical for downstream NLP based response and soft skill analysis.
- **Time bound:** Module development and initial testing will be completed during the first quarter of system implementation.

Objective 2: Implement NLP Based Response Evaluation Engine

- **Specific:** Develop a semantic analysis engine using SpaCy and Transformer based models to evaluate candidate answers for relevance, completeness, and alignment with job specific requirements.
- **Measurable:** Performance will be measured by similarity scores and correlation with human HR evaluations, targeting ≥0.8 correlation coefficient.
- **Achievable:** Pretrained transformer models can be adapted for context aware interview evaluation.
- **Relevant:** Addresses limitations of keyword based evaluation and ensure meaningful assessment of candidate responses.
- **Time bound:** Implementation and testing during the second quarter.

Objective 3: Design Communication & Behavioral Cue Analysis Module

- **Specific:** Implement a module to measure candidate communication patterns, including fluency, filler word usage, response time, and tone of speech.
- **Measurable:** Evaluate the system's ability to detect communication effectiveness compared to HR expert assessment.
- **Achievable:** NLP techniques, combined with audio feature extraction, can be used to quantify behavioral and communication indicators.
- **Relevant:** Enhances soft skill evaluation and provides insight into candidate interpersonal abilities.
- **Time bound:** Module development in the second and third quarters.

Objective 4: Develop Sentiment & Emotion Analysis System

- **Specific:** Implement emotion detection using sentiment analysis (VADER/TextBlob) to evaluate candidate confidence, positivity, and emotional tone.
- **Measurable:** Compare AI generated sentiment analysis with human observations to ensure the AI's interpretations align with human judgment.
- **Achievable:** Existing sentiment and emotion detection frameworks can be integrated and fine-tuned.
- **Relevant:** Addresses the research gap of incorporating emotional intelligence into recruitment evaluation.
- **Time bound:** Module development during the third quarter of the project.

Objective 5: Design Explainable AI Scoring & Reporting Framework

- **Specific:** Generate transparent Candidate Readiness Scores with evidence based explanations for each soft skill and communication metric.
- **Measurable:** Evaluation based on HR expert feedback on interpretability and usefulness of generated reports.
- **Achievable:** Techniques from explainable AI can be integrated with scoring models to produce interpretable reports.
- **Relevant:** Fills the transparency gap in AI based recruitment systems.
- **Time bound:** Development in the later stage of the project (fourth quarter).

Objective 6: Integrate System Components into a Unified Recruitment Platform

- **Specific:** Develop a web based platform integrating transcription, NLP response evaluation, communication analysis, sentiment assessment, scoring, and reporting modules.
- **Measurable:** Platform usability and end to end evaluation performance, supporting real time interview processing.
- **Achievable:** Existing web development frameworks and cloud infrastructure will enable smooth integration.
- **Relevant:** Ensures practicality and usability in real HR recruitment environments.
- **Time bound:** Integration to be completed before final system testing phase.

Objective 7: Evaluate System Performance and Reliability

- **Specific:** Conduct comparative testing using simulated and real interview datasets to assess system effectiveness.
- **Measurable:** Metrics include alignment with human resource evaluations, a meaningful reduction in the time required for assessments, and dependable precision in identifying soft skills.

- **Achievable:** Comparative evaluation of human judgments ensures realistic performance assessment.
- **Relevant:** Confirms system reliability, fairness, and applicability for practical recruitment.
- **Time bound:** Evaluation during the final quarter of the research project.

3. METHODOLOGY

This research adopts a Design Science Research (DSR) methodology to design, develop, and evaluate an intelligent AI based interview evaluation system. Design Science Research is appropriate for this study because it focuses on the creation of innovative technological artifacts that address practical real world problems. In the context of recruitment, the proposed system aims to enhance the traditional interview evaluation process by introducing automated analysis of candidate responses, communication behavior, and soft skills using artificial intelligence techniques.

3.1 Research Approach

The research follows an experimental system development approach, where the proposed AI based interview analysis module is designed, implemented, and evaluated through iterative development cycles. The methodology begins with identifying the problem and defining system requirements based on existing recruitment challenges. After the requirements are established, the system architecture is designed to integrate multiple AI components responsible for speech recognition, language processing, behavioral analysis, and credential validation. The development process includes implementing machine learning models, integrating natural language processing techniques, and building a prototype system capable of analyzing interview recordings. The system will then be tested using interview datasets to evaluate its performance and accuracy. The results obtained from the automated system will be compared with human evaluations to assess the effectiveness and reliability of the proposed solution.

3.2 Technical Methods and Algorithms

The proposed system incorporates several artificial intelligence and data processing techniques to analyze candidate interview responses and communication patterns. These techniques work together to generate an objective assessment of candidate soft skills and interview performance.

Speech Recognition

The interview evaluation process begins with the conversion of candidate speech into textual form. Interview recordings will be processed using the OpenAI Whisper speech recognition model, which is capable of accurately transcribing spoken language into text. This model supports multiple accents and noisy environments, making it suitable for real world interview scenarios.

Natural Language Processing

Once the audio is converted into text, Natural Language Processing (NLP) techniques are applied to analyze the content of candidate responses. NLP frameworks such as SpaCy and transformer based models will be used to evaluate semantic relevance, keyword alignment with job requirements, and the overall structure of responses. This analysis helps determine whether the candidate's answers demonstrate appropriate knowledge, understanding, and communication clarity.

Sentiment and Emotion Analysis

To evaluate the emotional tone and confidence level of candidates during interviews, sentiment analysis models such as VADER or TextBlob will be utilized. These models analyze textual features to identify emotional indicators such as positivity, neutrality, or negativity in candidate responses. The sentiment scores contribute to the overall assessment of a candidate's communication style and professionalism.

Communication Pattern Analysis

In addition to analyzing the content of responses, the system will also evaluate communication patterns using NLP based pattern recognition techniques. This includes identifying filler words, pauses, response length, and fluency patterns, which can provide insights into the candidate's confidence and clarity of speech. These behavioral indicators are important factors when assessing soft skills during interviews.

Certificate Validation

To improve trust and transparency in the recruitment process, the system also incorporates a certificate validation mechanism. Candidate credentials and certificates submitted during the application process will be verified using digital certificate metadata and verification APIs where available. This feature helps recruiters confirm the authenticity of candidate qualifications and reduce the risk of fraudulent credentials.

3.3 Tools and Platforms

The system will be implemented using a combination of modern programming tools, machine learning frameworks, and cloud technologies to ensure scalability and reliability.
The primary development environment will use Python, which provides extensive libraries and frameworks for artificial intelligence and natural language processing. The OpenAI Whisper model will be used for speech recognition tasks, while SpaCy and Hugging Face Transformers will support advanced language analysis.

The backend services will be implemented using Flask or FastAPI, enabling the system to expose APIs for interview processing and data management. A web based dashboard interface, potentially developed using React or similar web technologies, will provide recruiters with visualization tools to review candidate evaluations and reports.

For scalable processing and deployment, the system may utilize cloud computing platforms such as Amazon Web Services (AWS) or Google Cloud Platform (GCP). These platforms provide computational resources required for machine learning processing and enable the system to support multiple users simultaneously.

The selected technologies are widely used in AI based applications due to their reliability, scalability, and strong ecosystem support for machine learning and NLP development.

3.4 Data Collection and Ethical Considerations

The data required for this research will consist of interview recordings, candidate responses, and simulated recruitment datasets. These datasets will be obtained from publicly available sources and synthetic interview simulations designed to replicate realistic hiring scenarios.
To protect personal information and ensure responsible data usage, all datasets will be anonymized before processing. Any personally identifiable information will be removed to maintain data privacy and confidentiality.

Since the proposed system processes candidate speech data and potentially sensitive personal information, ethical considerations are an important aspect of this research. Ethical practices such as informed consent, secure data storage, and responsible AI usage will be strictly followed throughout the study. Ethical approval will be obtained before collecting any data involving human participants.

3.5 Validation Strategy and Performance Metrics

The performance and reliability of the proposed interview evaluation system will be assessed using several evaluation metrics. These metrics will measure the accuracy and effectiveness of different components within the system.

Speech recognition performance will be evaluated using the Word Error Rate (WER) metric, which measures the accuracy of speech to text transcription. The semantic analysis component will be evaluated by measuring the accuracy of semantic similarity scoring between candidate responses and expected interview answers.

The sentiment analysis module will be assessed using precision and classification accuracy to determine how effectively the system identifies emotional tone in candidate responses.
Finally, the overall evaluation generated by the AI system will be compared with manual assessments conducted by HR professionals. Statistical correlation analysis will be used to determine the level of agreement between automated scores and human evaluations. A strong correlation between these results would indicate that the system can provide reliable and objective interview assessments.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and Its Components

The proposed system is an AI-Based Interview Evaluation and Soft Skill Analysis module that operates as a component of the larger AI-Powered Intelligent Talent Acquisition System. The module is designed to automatically analyze candidate interview responses, communication patterns, emotional tone, and credential authenticity to generate an objective evaluation of candidate readiness.

Traditional interview evaluation processes rely heavily on subjective human judgment, which can lead to inconsistencies, bias, and inefficiencies. The proposed system addresses these limitations by integrating multiple Artificial Intelligence technologies to perform automated analysis of interview recordings and candidate responses.

The architecture consists of several interconnected components that work together to process interview data and generate evaluation results. These components include:

- **Interview Data Acquisition Module**
  This component collects candidate interview responses in the form of audio recordings or uploaded video/audio files. It acts as the entry point of the system.

- **Speech to Text Transcription Module**
  The recorded audio responses are processed using the OpenAI Whisper speech recognition model, which converts spoken interview responses into text transcripts. This step enables further natural language processing of candidate responses.

- **Natural Language Processing (NLP) Analysis Engine**
  The textual responses are analyzed using NLP techniques implemented through SpaCy and transformer-based models. This component evaluates semantic relevance, vocabulary usage, response clarity, and contextual alignment with interview questions.

- **Communication Pattern Analysis Module**

  This module evaluates communication behavior during the interview by analyzing features such as:

  - Response length
  - Speaking fluency
  - Filler word usage
  - Response timing and structure
  - These features provide insights into candidate confidence, articulation, and effectiveness in communication.

- **Sentiment and Emotion Analysis Module**

  Sentiment analysis tools such as TextBlob or VADER are used to detect emotional tone and attitude in candidate responses. This helps assess attributes such as confidence, positivity, and emotional stability.

- **Certificate Validation Module**

  Candidate certificates and qualifications submitted during the application process are verified through metadata inspection or external verification APIs to ensure authenticity.

- **AI-Based Scoring and Evaluation Engine**

  All outputs generated by the analysis modules are aggregated and processed by a scoring engine that calculates a Candidate Readiness Score. This score reflects communication skills, behavioral indicators, sentiment patterns, and response relevance.

- **Recruiter Dashboard and Reporting Module**

  The results are displayed in a dashboard interface that provides recruiters with detailed reports, visualizations, and explainable insights into candidate performance

By integrating these components, the system provides a comprehensive and automated interview evaluation framework that enhances objectivity, transparency, and efficiency of decision-making recruitment.

4.2 High-Level System Architecture Diagram

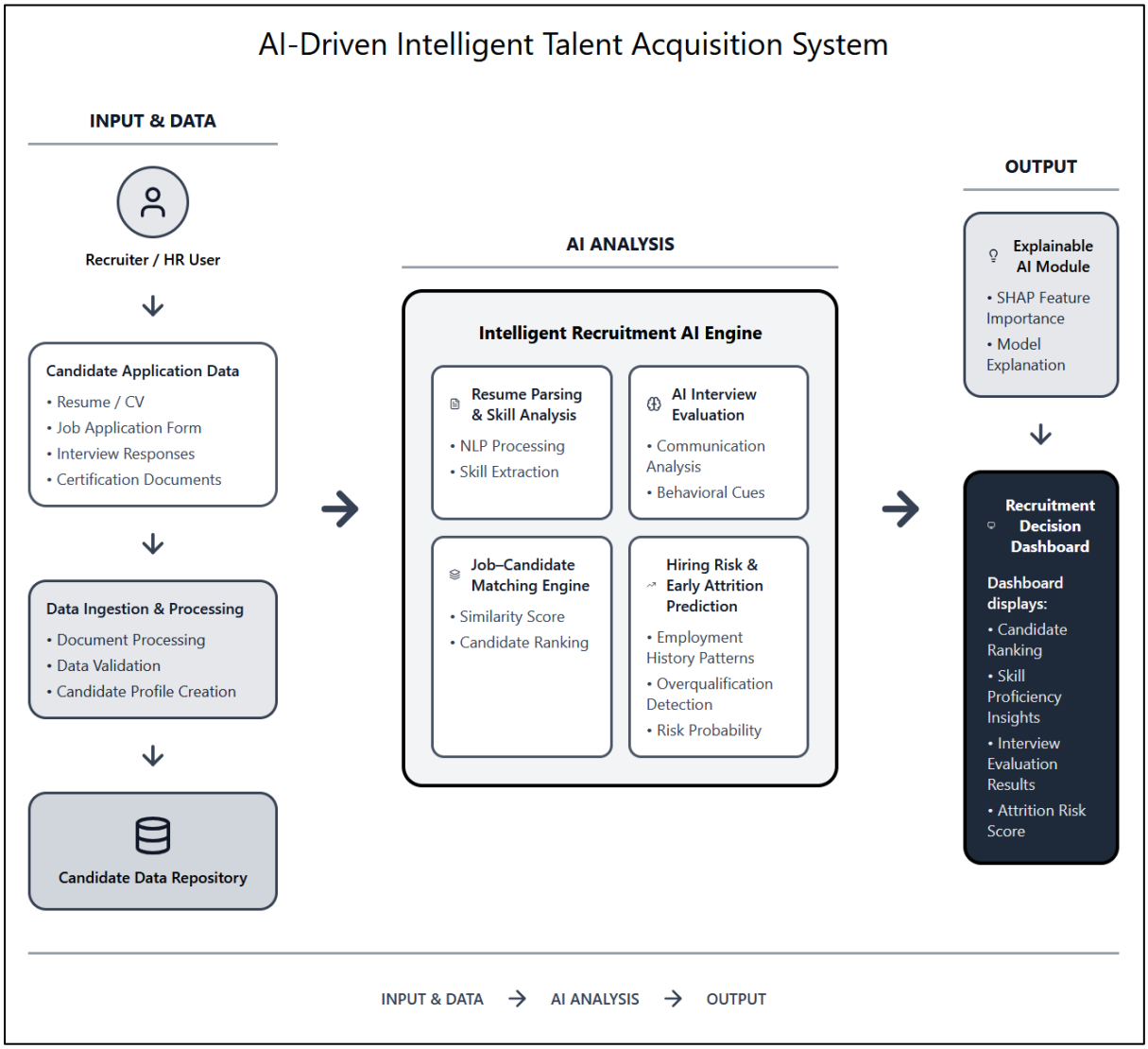


Fig.1. AI-Driven Intelligent Talent Acquisition System Architecture

Figure 1 illustrates the high-level architecture of the proposed AI-Driven Intelligent Recruitment Decision Platform. The system is organized into three main layers: Input & Data Layer, AI Analysis Layer, and Output & Decision Layer. The platform collects candidate information, processes data, applies artificial intelligence techniques for analysis, and delivers intelligent hiring insights to recruiters. The process begins with the Recruiter / HR User, who uploads candidate information such as resumes, job application forms, interview responses, and certification documents through the system interface. This data is processed by the Data Ingestion & Processing module, which performs document processing, validation, and conversion of raw inputs into structured candidate profiles. The processed data is then stored in the Candidate Data Repository, which acts as the centralized storage system for recruitment

data. The Intelligent Recruitment AI Engine serves as the core analytical component of the platform. It includes several AI modules such as Resume Parsing & Skill Analysis, AI Interview Evaluation, Job-Candidate Matching Engine, and Hiring Risk & Early Attrition Prediction. These modules analyze candidate qualifications, evaluate interview responses, calculate job-candidate similarity scores, and estimate the risk of early employee resignation. To ensure transparency, the system integrates an Explainable AI (XAI) module, which uses techniques such as SHAP to identify and explain the factors influencing AI predictions. Finally, the analyzed results are presented through the Recruitment Decision Dashboard, where recruiters can view candidate rankings, skill insights, interview evaluations, and attrition risk scores.
Overall, the system enables recruiters to make data-driven hiring decisions by transforming candidate information into meaningful insights through an integrated AI-powered recruitment workflow.

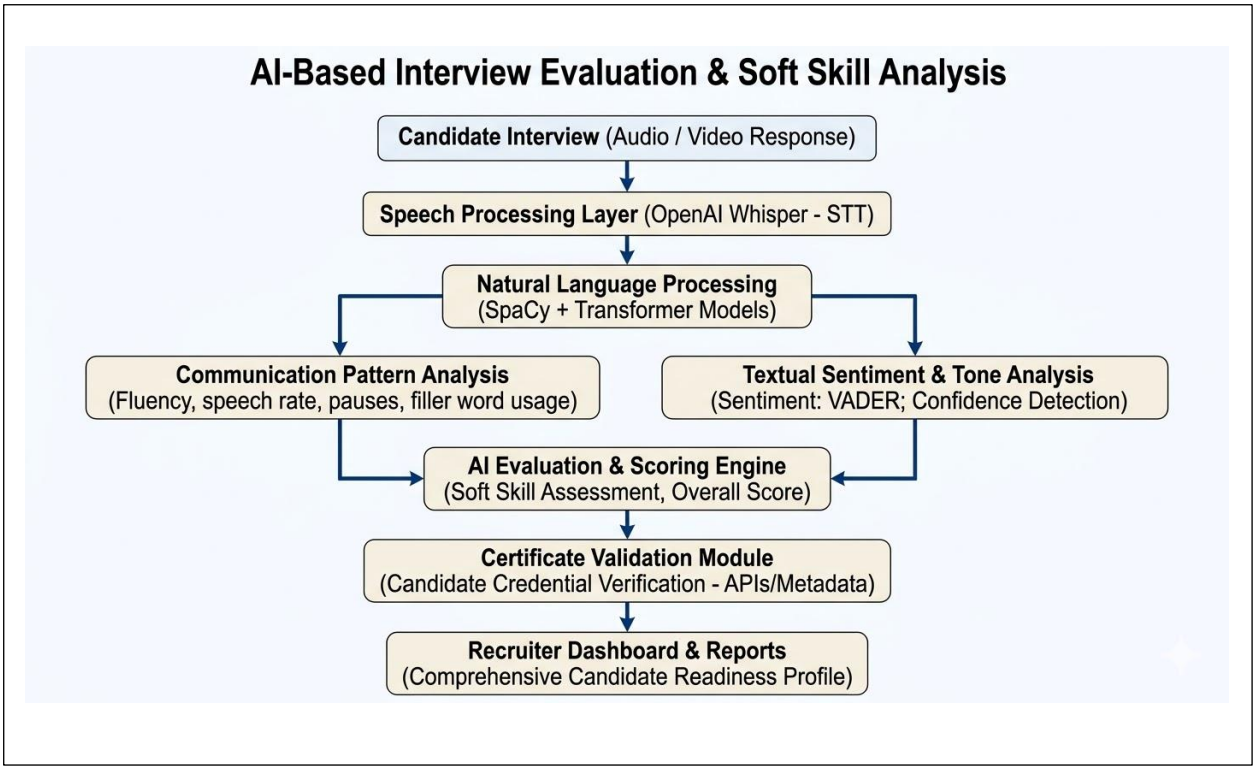


Fig.2. System Architecture for AI-Based Interview Evaluation & Soft Skill Analysis

Figure 2 illustrates the detailed architecture of the proposed AI-based interview evaluation and soft skill analysis module. The system is structured to process candidate interview data through multiple interconnected components, including speech-to-text transcription, natural language processing, communication pattern analysis, sentiment and emotion detection, and certificate validation. Each module performs a specialized function, contributing to a comprehensive assessment of candidate performance. The AI-based scoring engine integrates outputs from all

modules to generate a Candidate Readiness Score, which is then visualized through the recruiter dashboard. This architecture ensures a seamless workflow from data acquisition to evaluation, providing objective, transparent, and multi-dimensional insights into candidate communication skills, behavioral traits, and credential authenticity.

4.3 Component Interactions

The system components interact sequentially to process interview data and produce meaningful insights. When a candidate submits an interview response, the audio recording is first transmitted to the Speech-to-Text module. The Whisper model processes the audio stream and generates a text transcript.

The transcript is then passed to the Natural Language Processing module, where semantic analysis is performed to evaluate the relevance and quality of the candidate's responses. Simultaneously, the communication analysis module evaluates linguistic features such as fluency and filler word frequency.

The sentiment analysis module examines emotional tone and identifies patterns related to positivity, confidence, or uncertainty. These outputs are combined with communication features and semantic evaluation results within the AI scoring engine.

The certificate validation module operates in parallel by verifying the authenticity of candidate credentials submitted through the recruitment platform.

Finally, the scoring engine aggregates all evaluation metrics and produces a structured Candidate Readiness Profile, which is displayed through the recruiter dashboard.

4.4 Data Flow Explanation

- The system follows a structured data processing pipeline.

1. **Data Input Stage**
   Candidate interview responses are collected in audio or video format through the recruitment platform.

2. **Speech Processing Stage**
   The audio data is processed by the Whisper speech recognition model to generate accurate text transcripts.

3. **Text Processing Stage**
   The transcript is analyzed using NLP techniques to evaluate semantic meaning, response quality, and contextual relevance.

4. **Behavioral Analysis Stage**
   Communication features such as fluency, speech patterns, and filler word frequency are extracted and analyzed.

5. **Emotion Detection Stage**
   Sentiment analysis algorithms classify the emotional tone of candidate responses.

6. **Evaluation Stage**
   The AI scoring engine integrates outputs from all modules to generate an overall readiness score.

7. **Output Stage**
   Recruiters receive detailed reports, candidate rankings, and explainable insights through the dashboard.

- This data flow enables the system to transform raw interview recordings into structured evaluation metrics.

4.5 Scalability and Security Considerations

To support large-scale recruitment environments, the proposed system incorporates several scalability and security features.

Scalability

The architecture is designed using a modular microservice approach, allowing each AI component to operate independently. Cloud platforms such as AWS or Google Cloud can be used to deploy scalable services that handle multiple interview analyses simultaneously.

Containerization technologies such as Docker may be used to ensure flexible deployment and efficient resource utilization. This approach allows organizations to scale the system depending on recruitment demand.

Security

Since the system processes sensitive candidate information, strong security mechanisms are required.

All candidate data, including interview recordings and transcripts, will be encrypted during transmission and storage. Access control mechanisms will ensure that only authorized HR personnel can view evaluation results.

Secure authentication protocols and role-based access control will protect the system from unauthorized usage. Additionally, anonymization techniques will be applied during data processing to protect candidate privacy and comply with ethical research standards.

5. USER REQUIREMENTS

5.1 Stakeholders

- **Recruiters and HR Professionals**
  - Primary users who manage the hiring process.
  - They rely on the system to screen candidates objectively, evaluate soft skills, and access detailed interview reports.

- **Hiring Managers**
  - Oversee department level recruitment decisions.
  - They use the system's candidate readiness scores and behavioral insights to make informed hiring choices.

- **Job Candidates**
  - Individuals participating in interviews.
  - They interact with the system to record or submit responses, and benefit from fair, unbiased evaluation and possible feedback.

- **System Administrators**
  - Responsible for maintaining the system, managing user accounts, ensuring data security, and supporting cloud or server infrastructure.

5.2 Functional Requirements

- **FR1 - Candidate Response Submission**
  - The system shall allow candidates to record or upload interview answers for evaluation.

- **FR2 - Speech to Text Conversion**
  - Convert candidate audio responses into text using AI based speech recognition for further processing.

- **FR3 - Response Analysis**
  - Analyze candidate responses with NLP to evaluate semantic relevance, technical correctness, and clarity.

- **FR4 - Sentiment and Emotion Assessment**
  - Assess tone, confidence, and positivity in responses using sentiment analysis.

- **FR5 - Communication Pattern Evaluation**
  - Detect filler words, pauses, fluency, and response structure to assess communication effectiveness.

- **FR6 - Certificate Validation**
  - Automatically verify candidate credentials using digital certificate metadata or verification APIs.

- **FR7 - Candidate Scoring**
  - Generate a comprehensive readiness score combining technical, communication, and soft skill metrics.

- **FR8 - Reporting and Dashboard**
  - Provide recruiters with detailed visual reports, including explanations of scores and behavioral insights.

5.3 Non-Functional Requirements

Security - All candidate data, including interview recordings, transcripts, and personal information, must be encrypted and securely stored to prevent unauthorized access, data breaches, or misuse. This ensures privacy and compliance with data protection regulations.

Performance - The system should process interviews efficiently, converting audio to text, analyzing responses, and generating reports within a reasonable timeframe. Fast processing ensures recruiters can evaluate candidates without unnecessary delays.

Scalability - The system must handle multiple interviews simultaneously without performance degradation. This allows organizations to use the platform for large scale recruitment events or high application volumes.

Usability - Recruiters and HR staff should be able to easily access and navigate candidate evaluation reports, dashboards, and analytics. A user friendly interface ensures effective decision-making and reduces training requirements.

Reliability - The system must maintain consistent accuracy and uptime, ensuring that interview analysis results are dependable and that the platform is available whenever needed. Reliable performance builds trust in AI driven recruitment decisions.

6. COMMERCIALIZATION PLAN

The proposed AI Based Interview Evaluation and Soft Skill Analysis system targets a broad range of potential customers, including small and medium enterprises, corporate HR departments, recruitment agencies, and online hiring platforms. These stakeholders often face challenges in efficiently evaluating large volumes of candidates while maintaining fairness and minimizing bias, making the system highly relevant to their operations.

The value proposition of the system lies in its ability to provide objective, data driven evaluations of candidates during interviews, extending assessment beyond resumes to include communication patterns, emotional cues, and soft skills. By automating these analyses, the system reduces recruiter workload, improves hiring efficiency, and supports fairer and more transparent recruitment practices.

The system's revenue model is designed to be flexible and scalable. Organizations can adopt Software as a Service subscription model, purchase enterprise licenses for unlimited access, or pay per interview analysis depending on their recruitment volume. The cost estimation accounts for system development, cloud infrastructure for scalable processing, and ongoing maintenance and support. A tiered pricing strategy based on the number of interviews analyzed per month ensures accessibility for small organizations while offering comprehensive options for larger enterprises.

The proposed system offers several competitive advantages over existing recruitment technologies. These include emotion aware interview analysis, soft skill evaluation beyond resumes, explainable AI scoring for transparency, and integrated certificate validation to ensure candidate credibility. These features differentiate the system from traditional Applicant Tracking Systems and basic video interview platforms.

Finally, regarding intellectual property considerations, the AI interview evaluation algorithm, scoring framework, and integrated analytical components may be protected through software copyright and potential patent registration, safeguarding the innovation and ensuring commercial exclusivity.

7. BUDGET AND JUSTIFICATION

Table 2: Estimated expenses of the proposed system

Category	Requirement / Resource	Description	Estimated Cost (LKR)
Hardware	External Backup Storage (SSD)	Storage for interview recordings, datasets, and trained models	18,000
Hardware	Audio Recording Equipment	Microphone & headset for collecting interview data	12,000
Hardware	Internet/Data Costs	Required for cloud access, uploads, and testing	10,000
Software	Programming Environment (Python, VS Code)	Development tools for backend and AI modules	0 (Open Source)
Software	AI & NLP Frameworks	Whisper, SpaCy, HuggingFace, TextBlob	0 (Open Source)
Software	Data Processing Libraries	Pandas, NumPy, Scikit-learn	0 (Open Source)
Software	Version Control (GitHub)	Code management and collaboration	0 (Free Tier)
Cloud Services	Cloud Computing Instance	GPU/CPU for AI model execution and testing	35,000
Cloud Services	Cloud Storage & Database	Storage for transcripts, user data, logs	22,000
Cloud Services	API Usage / AI Services	Optional APIs (speech/NLP if needed)	8,000
Security	Basic Security & Authentication	Token-based authentication, encryption setup	5,000
Testing & Deployment	System Testing & Debugging	Testing tools, bug fixing, optimization	5,000
Documentation	Report Printing & Materials	Printing final reports and documentation	3,000
Human Effort	System Development	Design, coding, testing (academic work)	0
Total Estimated Cost			**118,000 LKR**

The proposed budget focuses on providing the necessary resources to develop and evaluate an AI-based interview evaluation system while maintaining cost efficiency. Since a personal computer is already available, the budget prioritizes storage, cloud computing, and operational expenses. External backup storage is included to manage large volumes of interview recordings, processed datasets, and trained machine learning models. As audio and transcription data require significant space, this ensures secure and reliable data storage. Audio recording equipment such as a microphone and headset is essential for collecting realistic interview responses, enabling accurate testing of speech recognition and communication analysis features. Internet and data costs are also considered, as continuous connectivity is required for accessing cloud platforms, uploading datasets, and running remote AI services. The software component of the system relies entirely on open-source tools, including Python, Visual Studio Code, and widely used AI and NLP frameworks such as Whisper, SpaCy, HuggingFace Transformers, and TextBlob. Data processing libraries like Pandas, NumPy, and Scikit-learn are used for preprocessing and model evaluation. Version control platforms such as GitHub are used for code management and collaboration at no additional cost. Cloud services play a critical role in the system. Cloud computing resources provide scalable processing power for running AI models and handling multiple user requests efficiently. Cloud storage and databases ensure secure management of candidate information, interview recordings, and analysis results. Additionally, a small allocation is included for API usage in case external AI services are required during development. Security is also considered by allocating resources for implementing authentication mechanisms and data protection measures, ensuring safe handling of sensitive candidate information. Testing and deployment costs are included to support debugging, system optimization, and ensuring reliability before final delivery. Documentation costs cover report preparation and printing required for academic submission. Human effort is not included in the financial estimate, as system design, development, and testing are carried out as part of the academic project. Overall, the total estimated cost of **118,000 LKR** provides a realistic and well-rounded budget to successfully develop, test, and deploy a functional AI-based interview evaluation system using modern cloud technologies and open-source tools.

8. WORK BREAKDOWN STRUCTURE (WBS)

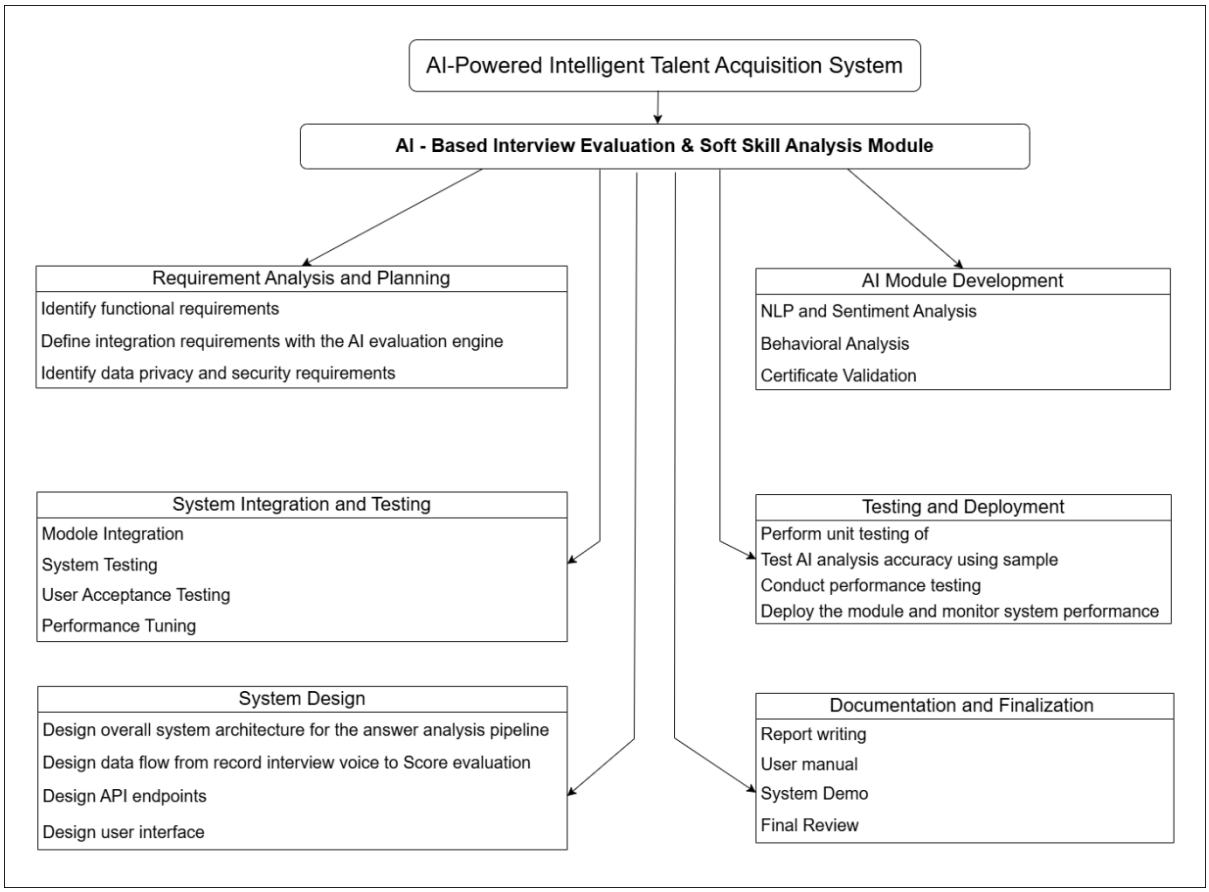


Fig.3. Work Breakdown Structure

Figure 3 Work Breakdown Structure (WBS) diagram provides a hierarchical representation of the project tasks. It visually organizes the research and development activities into manageable components, showing dependencies and timelines for each task. The WBS ensures balanced workload distribution, facilitates tracking of project progress, and helps align individual responsibilities with overall project objectives for the AI-Based Interview Evaluation and Soft Skill Analysis system.

9. GANTT CHART

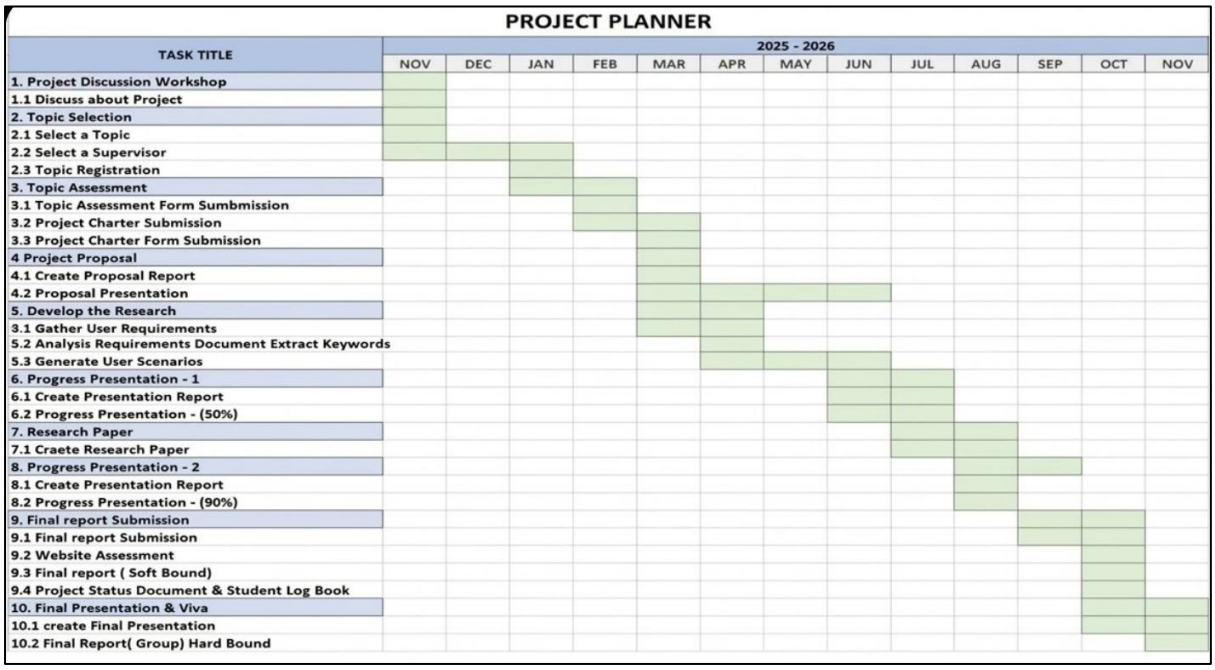


Fig.4. Gantt Chart

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11.APPENDICES

Appendices 1:Risk analysis

Risk	Description	Impact	Likelihood
Speech Recognition Errors	Accents, background noise, or poor audio quality may reduce the accuracy of speech-to-text transcription during interview analysis.	Medium	Medium
NLP Misinterpretation	Natural Language Processing models may incorrectly interpret candidate responses, affecting soft skill evaluation accuracy.	Medium	Medium
Fake Certificate Submission	Candidates may upload forged or manipulated certificates to gain unfair advantages.	High	Medium
Data Privacy Breaches	Candidate interview recordings, resumes, and certificates contain sensitive personal data that could be exposed if security is weak.	High	Low
System Performance Issues	Processing large numbers of interview recordings may slow down the system or increase response time.	Medium	Medium
Model Overfitting	AI models trained on limited datasets may not generalize well to real-world interviews.	Medium	Medium
Integration Failures	Integration between different modules such as speech recognition, NLP analysis, and certificate validation may cause system failures.	Medium	Low
Ethical Concerns in AI Decision Making	Over-reliance on AI-based evaluation could lead to unfair hiring decisions if not properly monitored.	High	Low

Appendices 2 : Flow chart

